The California Solar Initiative

by Cecelia Barros
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For an increasing number of Californians, renewable energy is an obvious choice. Harnessing the earth's own inexhaustible energy—whether from the sun, wind, biomass, or other renewable sources—can reduce dependence on fossil fuels and provide clean, affordable electricity. California has made a bold decision to invest in solar energy and move the state toward a cleaner energy future. As part of Governor Arnold Schwarzenegger's Million Solar Roofs program, California has set a goal to create 3,000 megawatts of new, solar-produced electricity by 2017.

PG&E

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On August 21, 2006, Governor Schwarzenegger signed Senate Bill 1 (SB 1), which directs the California Public Utilities Commission (CPUC) and the California Energy Commission (CEC) to implement the California Solar Initiative (CSI) program for the period 2007–2016, consistent with specific requirements and budget limits as set forth in the legislation. The CPUC, through the CSI, will be providing more than $2 billion in incentives over the next decade for existing residential homes and existing and new commercial, industrial, and agricultural properties. The CEC is managing a ten-year, $350 million program to encourage solar installations in new-home construction through its New Solar Homes Partnership (NSHP). The overall goal of the CSI is to help lower the cost of solar systems for consumers and build a self-sustaining solar market.

Energy Efficiency Audits

On March 2, 2006, the CPUC opened a proceeding to develop rules and procedures for the CSI. The procedures for the first time include energy efficiency prerequisites that must be met in order to receive a rebate on the cost of a solar system, and base rebates on the performance of a new solar system, rather than simply on its size (see "Why Energy Efficiency?").

As of January 1, 2007, Pacific Gas and Electric Company (PG&E) will be the administrator and implementer of this program to customers within our service territory. PG&E customers who are interested in participating in the CSI must contact PG&E for a solar incentive application. Also beginning January 1, 2007, all existing residential and commercial customers must have an energy efficiency audit conducted on their existing home or building if they choose to apply for a solar incentive. The acceptable audit protocols will consist of an online audit or a telephone audit provided by the utilities. The utilities may make other audit tools available for customers in addition to the online and phone audits. For PG&E residential customers, a mail audit will also be allowed. For commercial customers, there are also additional tools available that are based on the energy usage of the customer's building or facility. Customers will be responsible for submitting a copy of a completed energy efficiency audit to the utility with their solar incentive application. At the minimum, the online and phone audit requirements would also apply to nonutility providers and would be at the expense of the customer.

PG&E will be encouraging customers who complete an energy audit to change out and upgrade older, inefficient systems and appliances before adding a solar system. Customers who do so may also be eligible to earn energy efficiency rebates from PG&E, reducing their overall energy usage and increasing the comfort of their home. This may also help them to properly size the solar system for their home, which may reduce the size and cost of the solar system.

AUDIT EXEMPTIONS

An existing home or building may be exempt from the audit requirement if it meets one of the following three criteria. A copy of the documentation must be submitted with the customer's solar incentive application.

1. Having an acceptable energy audit report during the past three years. Examples of acceptable energy audit reports include a copy of an energy audit report summary completed through a customer's local utility company, a home inspection report from an independent vendor or consultant that includes an energy efficiency analysis, or a Home Energy Rating System (HERS) summary from a certified HERS rater. (Note: Residential Energy Services Network (RESNET) is now working on defining an audit versus a rating. A rating is not the same thing as an audit, since it rates the real estate, not the occupants. An audit uses bill history, so it is like the occupants' fingerprint; it is related to actual consumption instead of modeled consumption. RESNET may argue that a rating, though it is not an audit, fulfills this requirement.)

2. Having proof of Title 24 energy efficiency compliance within the past three years. For a residential building, form CF-1R, which was used to demonstrate Title 24 compliance with the 2001 or 2005 Energy Efficiency standards, and was generated on or after January 1, 2003, will be sufficient to demonstrate compliance. For a commercial building, one or more of the Certificate of Compliance forms listed in Table 1 could be used to demonstrate Title

Why Energy Efficiency?

Just as it makes no sense for electric utilities to build new plants to fuel inefficient customer equipment, so it makes no sense to size a personal solar power plant based on inefficient lighting, appliances, heating, cooling, water heating, and electronics. In fact, state policy places energy efficiency at the top of the loading order, ahead of new generation systems. This is for the right reason! Similarly, for consumers who want to zero out carbon emissions, employing NegaWatts—the kilowatts saved by energy efficiency projects—is typically the best first step. Though energy efficiency projects are not as glamorous as solar panels, they really are just as green. So take full advantage of the energy audit services of your utility; consider a diagnostic test-based home rating or home performance check by a Home Performance with Energy Star contractor.

Here is an example of just how important it is to implement efficiency first. We compared two scenarios aimed at zeroing out a San Francisco customer's electrical consumption without producing any excess electricity. The first was to install a solar PV system that would produce as much electricity as the customer's current yearly usage. The second scenario involved first replacing the customer's older fixtures and appliances with more energy-efficient ones, and only then installing a now-smaller PV system to produce as much electricity as the customer would now consume on an annual basis. The second scenario would cost $10,450 less to implement than the first one, and would still result in a zero carbon footprint for this customer.

Scenario One: Size Solar PV to existing electricity consumption of 7,735 kWh per year

- System size: 6.0 kW
- Net cost after incentives: $34,720
- Electrical production: 7,735 kWh
- Electrical consumption carbon footprint = *ZERO*

Scenario Two: Reduce Solar PV by 33% by implementing basic, reliable energy efficiency measures

- System size: 3.9 kW
- Net cost of solar after incentives: $22,170
- Electrical production: 4,826 kWh
- Approximate costs of implementing energy efficiency measures:
  - $500 CFL fixtures replace incandescents
  - $100 CFLs screw-in lighting in recessed fixtures and other lamps without dimmers
  - $900 Consortium for Energy Efficiency (CEE) top tier clothes washer
  - $500 CEE top tier dishwasher
  - $100 power strips to positively turn off electronic devices or motion sensor power strips for home office equipment
  - Total: $2,100 Energy efficiency investment (full cost not incremental)
  - Energy efficiency savings: 33% or 2,575 NegaWatts
  - Total cost of solar and efficiency: $24,270
  - Savings compared to Scenario One: $10,450
  - Electrical consumption carbon footprint = *ZERO*

(Note: this example is based on the experience of a typical household in San Francisco; it assumes somewhat limited solar production and is not optimized for time-of-use rates. An example based on conditions in California's hotter Central Valley region would improve both paybacks.)

POWERLIGHT CORP

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Table 1. Acceptable Compliance Forms

Envelope	Mechanical	Lighting	Outdoor Lighting
ENV-1-C	MECH- 1-C	LTG-1-C	OLTG-1-C

24 compliance with the 2001 or 2005 Energy Efficiency standards, provided that it was generated on or after January 1, 2003.

Only compliance forms completed by persons who are Certified Energy Plans Examiners (CEPE) by the California Association of Building Energy Consultants (CABEC) will be accepted. These forms must be generated by one of the Energy Commission's approved Title 24 software programs—either Micropas or Energy Pro.

3. Having one of two national certifications of energy efficiency, either LEED or Energy Star. To meet this criterion, the customer must submit a copy of a certificate or other documentation from LEED or Energy Star that contains the address of the building and the date of certification.

NEW CONSTRUCTION

The requirements are slightly different for new-construction projects. Residential new-construction projects—single-family homes, custom homes, and multifamily buildings—are currently handled by the CEC under the NSHP. A very important feature of the NSHP is that builders are required to exceed Title 24 performance by 15% (not including the PV system) to qualify for the rebate program. PG&E will be working with the other statewide utilities to develop training for HERS raters who will be required to inspect solar installations in new homes as a requirement under the NSHP. PG&E anticipates that these HERS raters will also be used as solar inspectors for installation on existing residential homes.

For commercial new construction, building owners who would like to participate in the solar incentive program must submit copies of their current Title 24 documentation, including one or more of the Certificate of Compliance forms listed in Table 1 that were used to demonstrate Title 24 compliance with the 2005 Energy Efficiency standards, which went into effect as of October 1, 2005.

Only compliance forms completed by persons who are Certified Energy Plans Examiners (CEPE) by the California Association of Building Energy Consultants (CABEC) will be accepted. These forms must be generated by one of the Energy Commission's approved Title 24 software programs—either Micropas or Energy Pro.

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For more information:

To learn more about the CSI requirements for existing buildings, or to obtain a solar rebate application, contact PG&E at www.pge.com/about_us/environment/solar/.

Contact the CEC for program requirements and for applications related to new construction at www.gosolarcalifornia.ca.gov.

To sign up for or to learn more about any one of PG&E's 200 classes on energy efficiency for design, construction, and building management professionals, visit www.pge.com/energyclasses.

While many of the classes are held in PG&E's classroom facilities in Stockton or San Francisco, trainings will also be held at 25 other locations across Northern California this spring. In addition, 14 classes will be simulcast over the Web, available to California rate payers or others with energy efficiency projects in California. Use the search function on the PG&E Web site to find the location nearest to you.